MACHINE LEARNING & DEEP LEARNING

WITH TENSORFLOW

By Taufiq Huda

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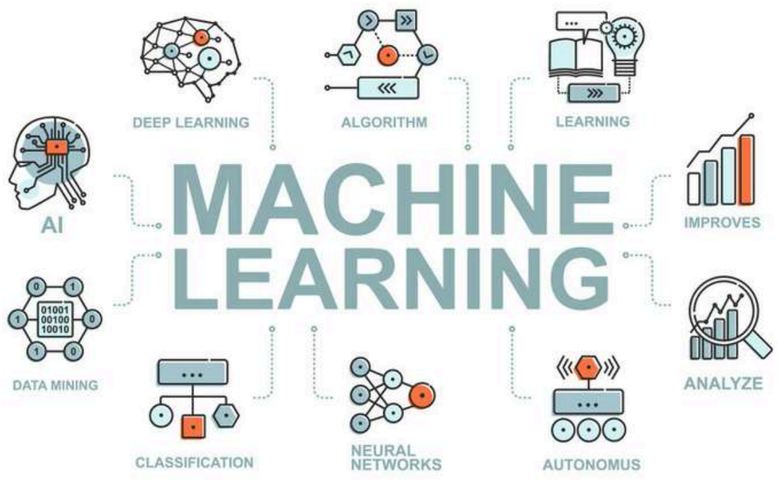
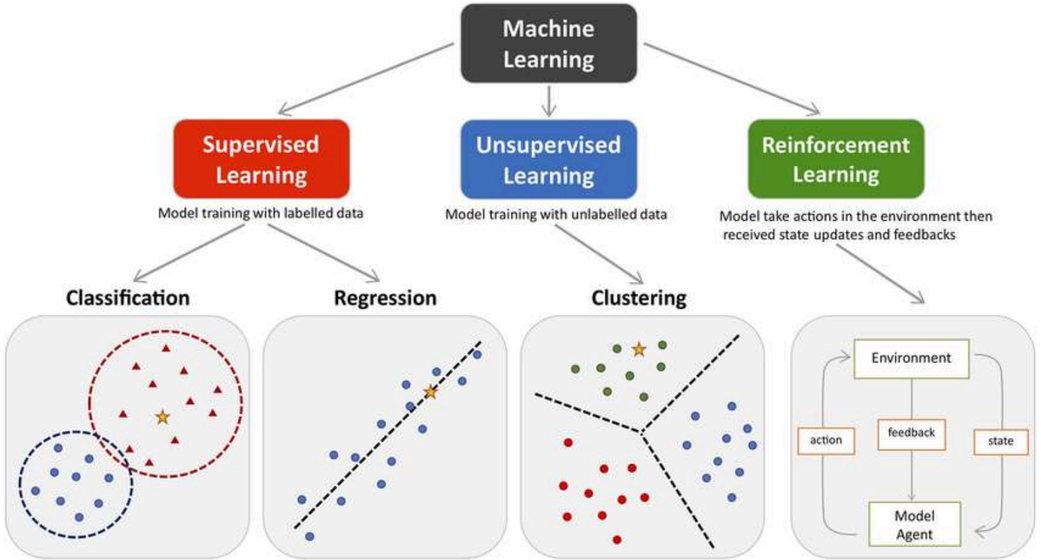
MACHINE LEARNING

a subset of artificial intelligence, is transforming the way we interact with technology. By enabling computers to learn from data and improve their performance over time without being explicitly programmed. In healthcare, it aids in diagnosing diseases. In finance, it helps detect fraudulent activities and assess credit risks.



Machine Learning usage for



Type of Machine Learning

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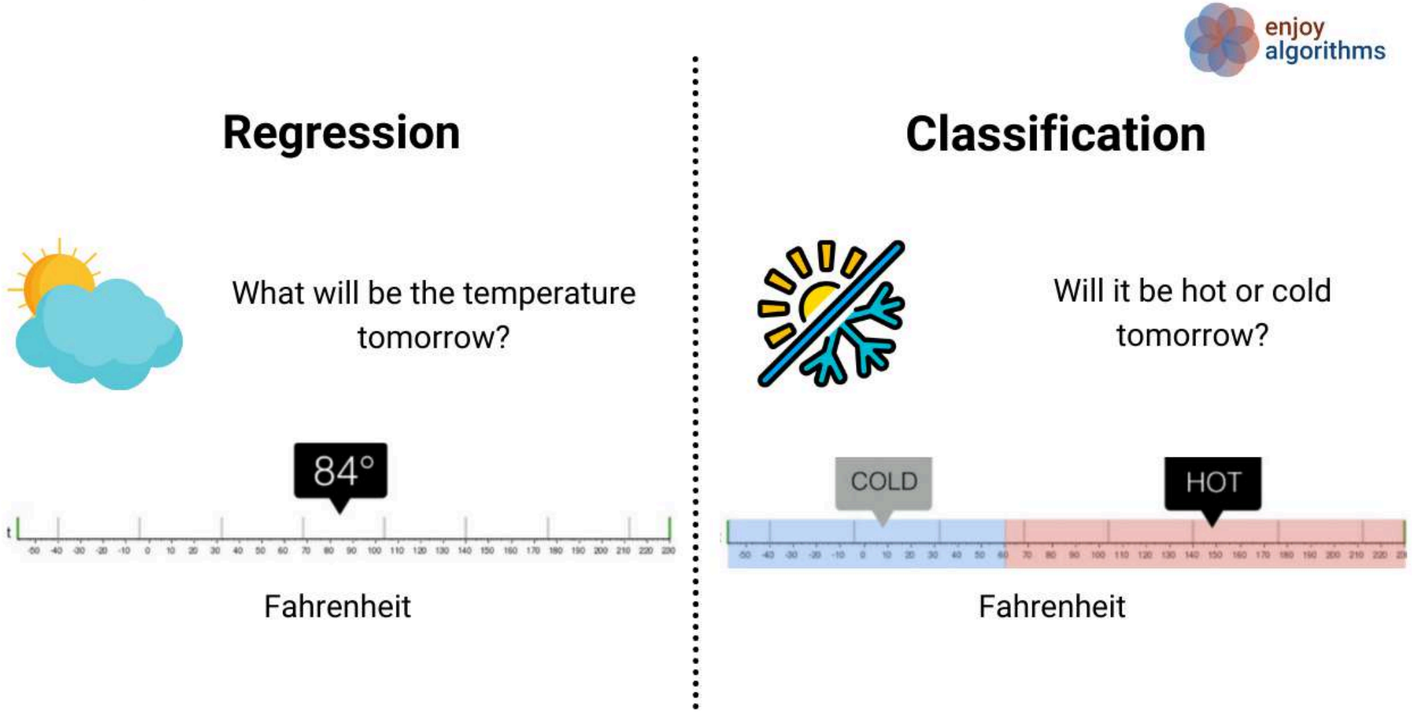
SUPERVISED LEARNING

is a type of machine learning where an algorithm is trained on a labeled dataset. This means that each training example is paired with an output label, allowing the model to learn the relationship between input data and the desired output. By iterating over the data, the model adjusts its parameters to minimize errors in its predictions.



Type of Supervised Learning

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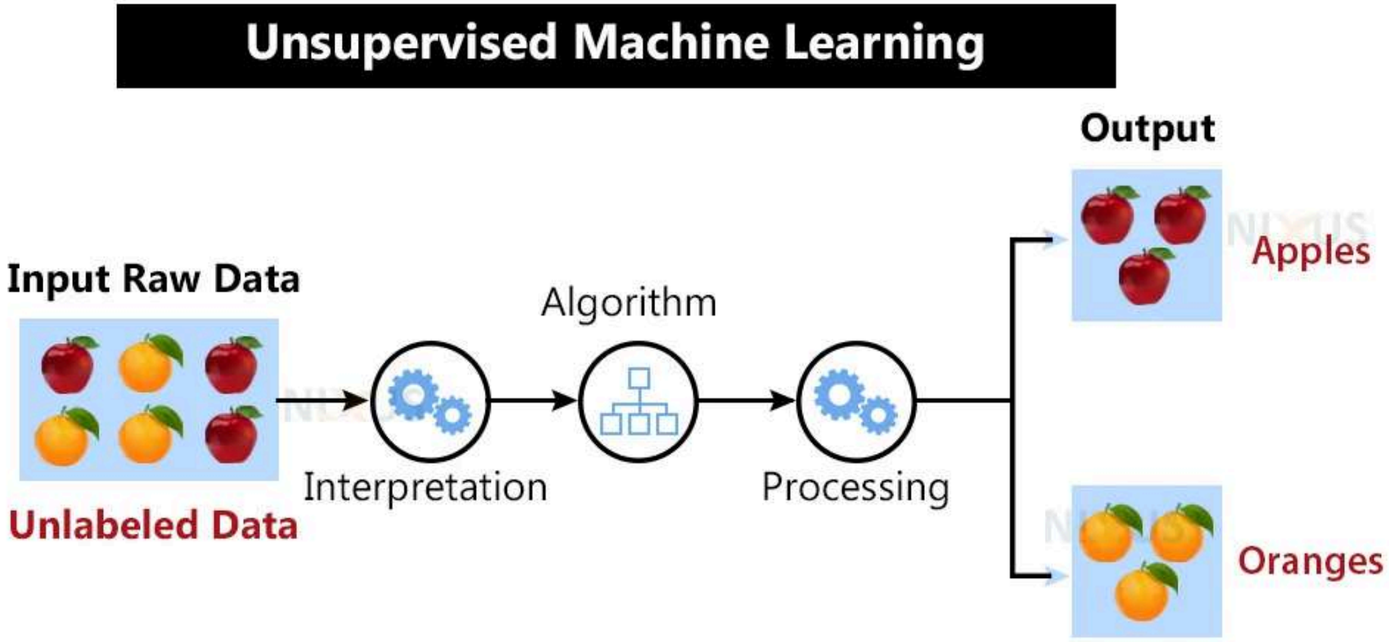
UNSUPERVISED LEARNING

a type of machine learning where the model is trained on data without explicit labels or guidance. This approach allows the model to discover hidden patterns, structures, or features in the data on its own.



Unsupervised Learning

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MACHINE LEARNING IMPLEMENTATION

Oil, Gas, and other fuels Price Prediction

# open Market opening price	# high Highest price during the trading day	# low Lowest price during the trading day.	# close Market closing price.	# volume Number of contracts traded during the day.
-14 … 146	0.51 … 147	-40.3 … 144	-37.6 … 146	0 … 2.29m
31.950000762939453	32.79999923706055	31.950000762939453	32.04999923706055	79385
31.899999618530273	32.2400016784668	31.399999618530273	31.6299991607666	72978
31.700000762939453	32.099998474121094	31.31999969482422	32.04999923706055	44601
32.040000915527344	32.91999816894531	31.860000610351562	32.869998931884766	46770
32.81999969482422	33.029998779296875	32.560001373291016	32.720001220703125	49131
32.75	33.400001525878906	32.099998474121094	33.400001525878906	79214
33.25	33.70000076293945	32.970001220703125	33.099998474121094	56895
33.04999923706055	33.45000076293945	32.75	33.380001068115234	45869

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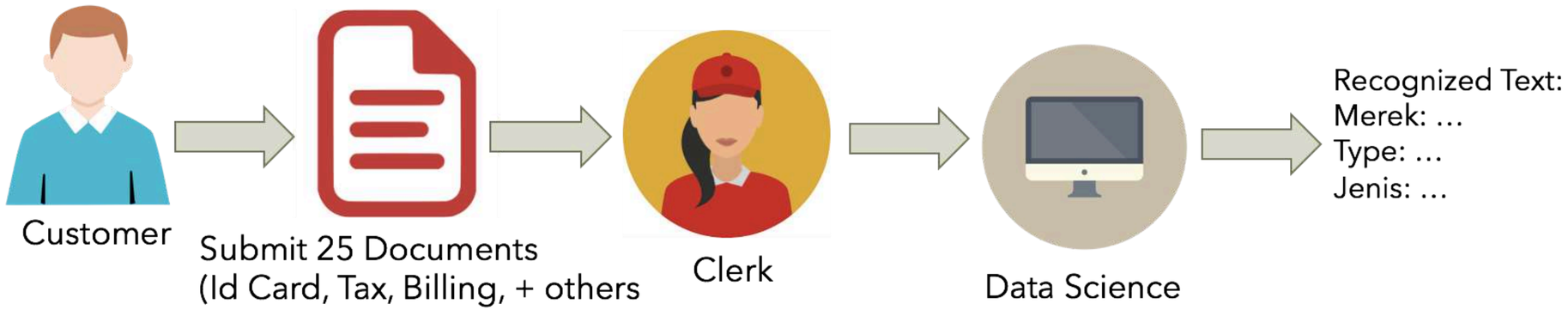
MACHINE LEARNING IMPLEMENTATION

OCR for Document Recognition in Vehicle Sales Company. Before AI implemented in vehicle sales company, for each transaction input required almost 1-2 hours for inputing data by clerk, hence the automated OCR Document system put in transaction that reduce inputing time to 5 minutes



Citizen id card detection using image processing and optical character recognition
W Satyawan, MO Pratama, R Jannati, G Muhammad... - Journal of Physics: Conference Series, 2019

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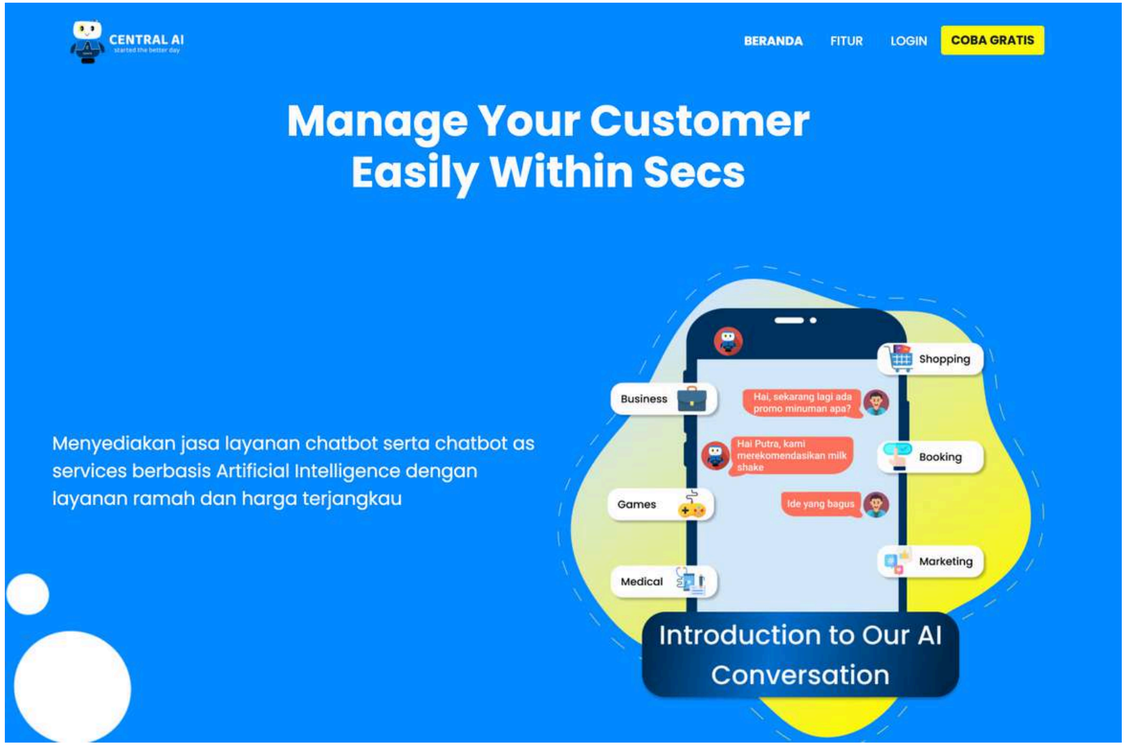
MACHINE LEARNING IMPLEMENTATION

Chatbot



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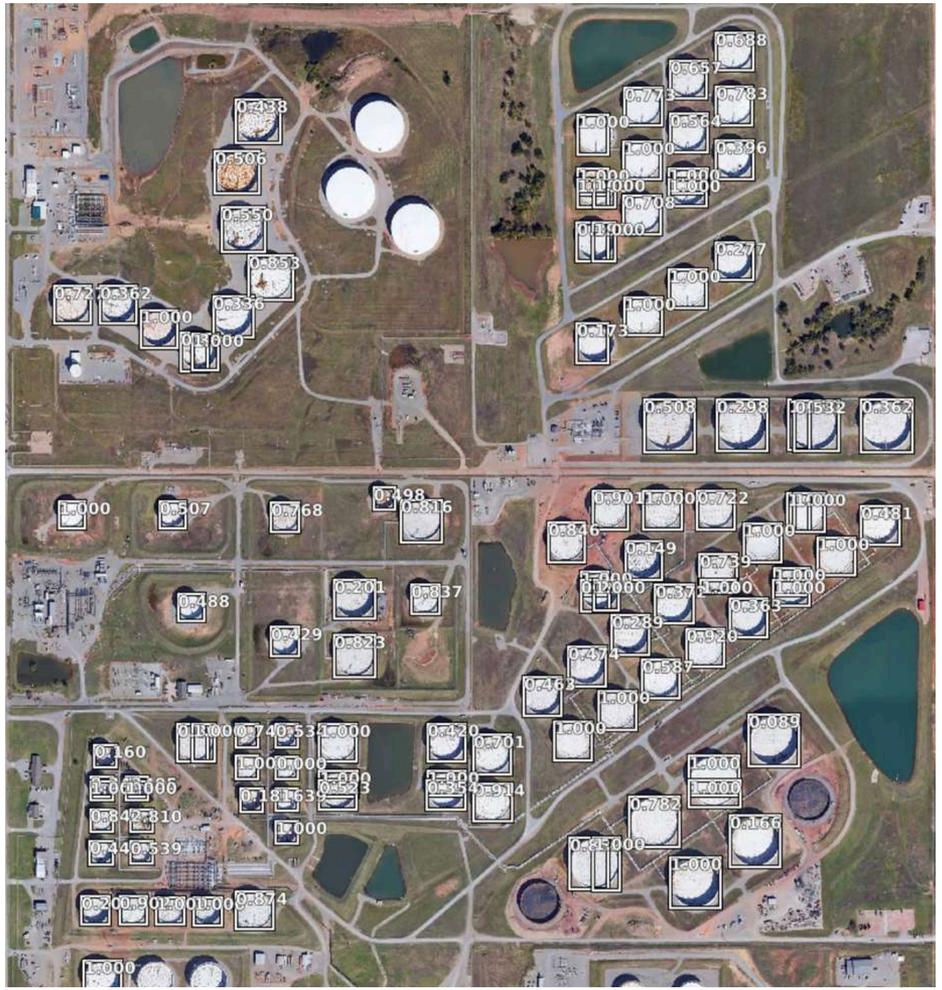
MACHINE LEARNING IMPLEMENTATION

Oil Tank Volume Estimation. Tank volume estimation is possible because oil is typically stored in floating head tanks. This particular tank type has a head that sits directly on top of the crude oil to prevent buildup of fumes. As a result, the height of the tank head rises and falls with the volume of oil in the tank. The relative sizes of the exterior shadow cast by the tank itself and the interior shadow cast by the height of the tank head can be used to estimate the tank volume.



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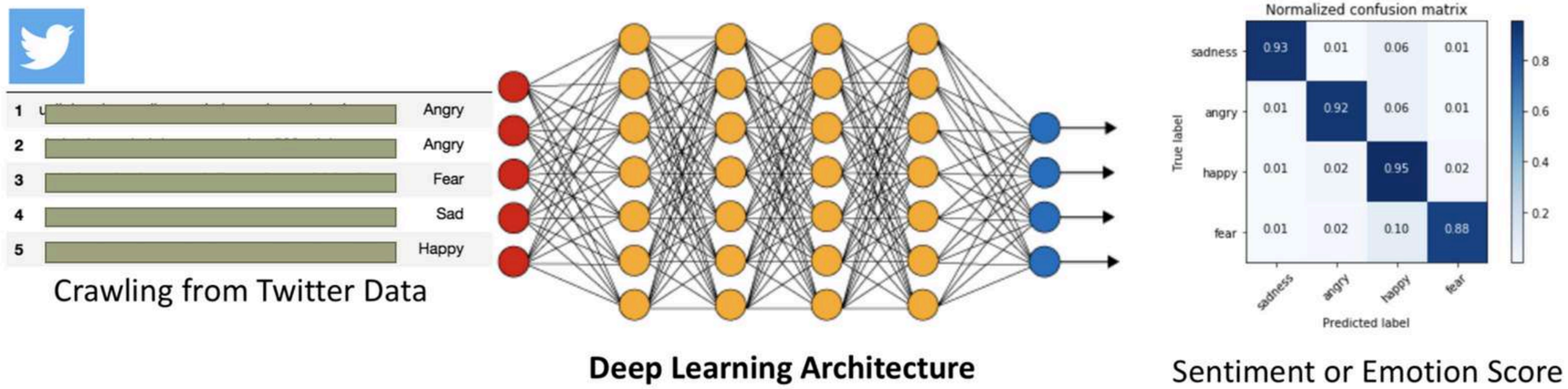
MACHINE LEARNING IMPLEMENTATION

Product Review or Sentiment Analysis can be done by using Data Gathering from Social Media then the model can be created to predict mood or sentiment of some products.

Sentiment analysis is useful for detecting emotion or sentiment from topics or products. Data is gathered from social media like Twitter, then Machine Learning is performed to detect emotion or sentiment in topics or products.



The sentiment analysis of Indonesia commuter line using machine learning based on twitter data
MO Pratama, W Satyawan, R Jannati, B Pamungkas… - Journal of Physics: Conference Series, 2019

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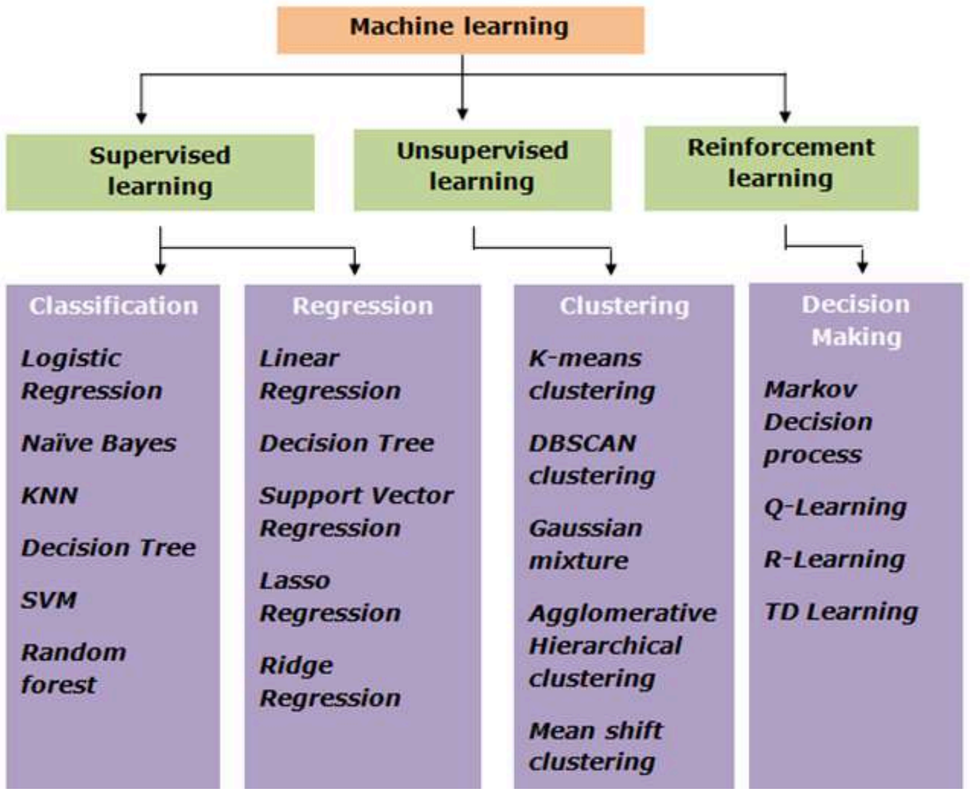
ALGORITHM



Machine Learning Common Algorithm

- Bagging (Bootstrap Aggregating)
- Random Forest
- Boosting
- Stacking
- Voting
- Blending

Ensemble Learning

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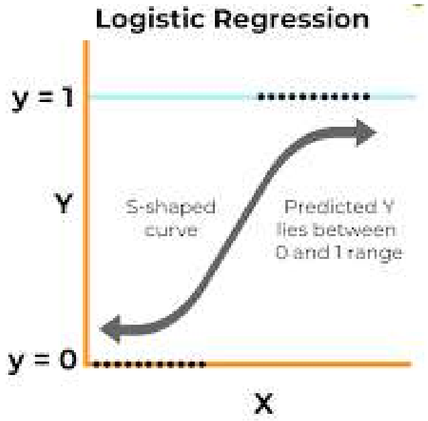
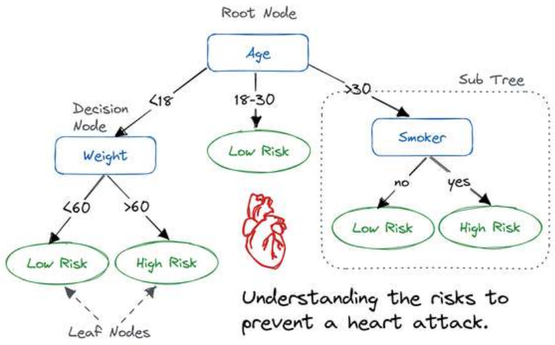
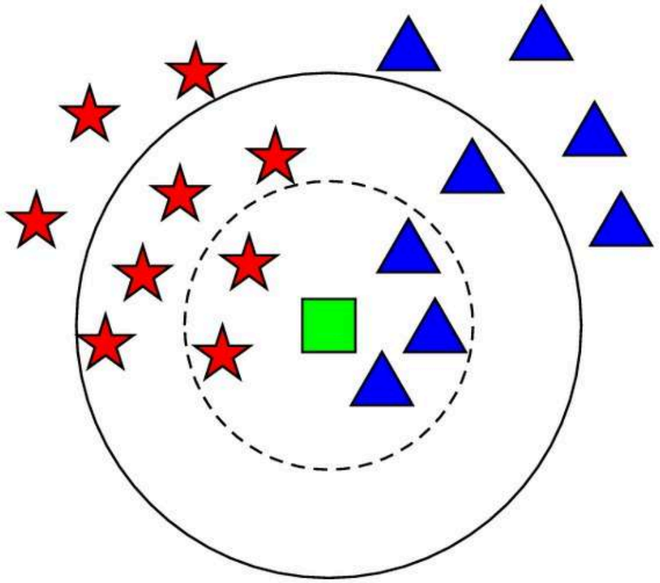
CLASSIFICATION



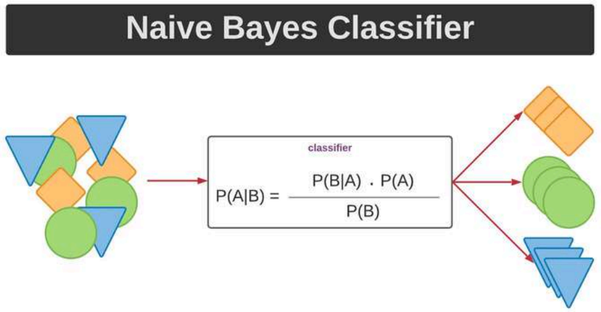
Logistic Regression



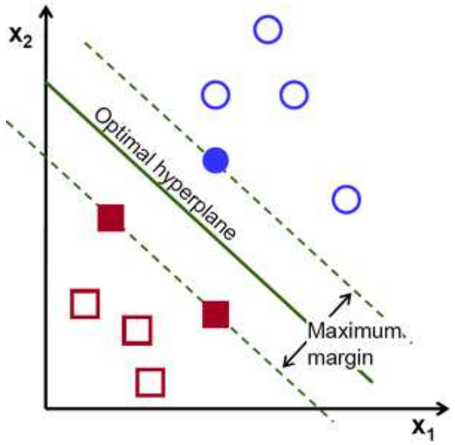
Decision Tree



K-Nearest Neighbors



Naive Bayes



Support Vector Machines

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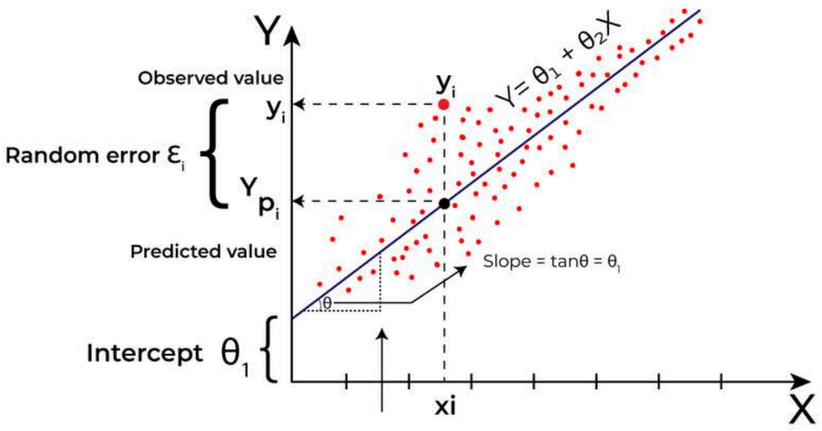
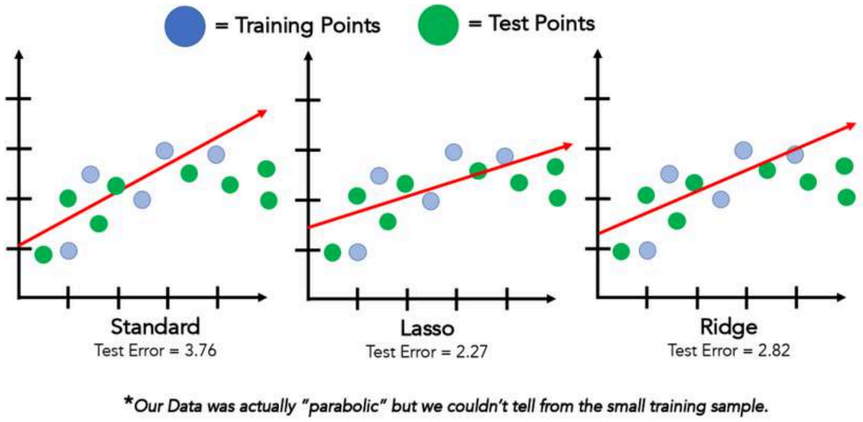
REGRESSION



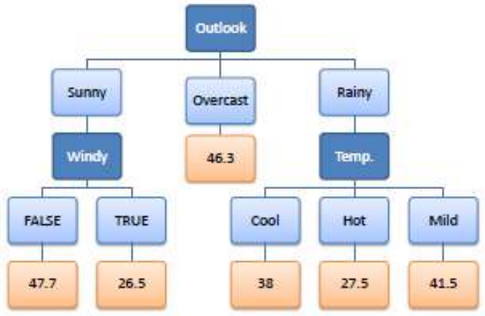
Linear Regression



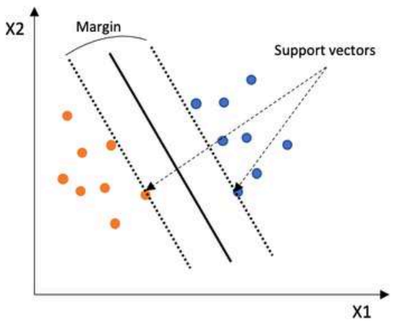
Lasso & Ridge Regression

Predictors | Target

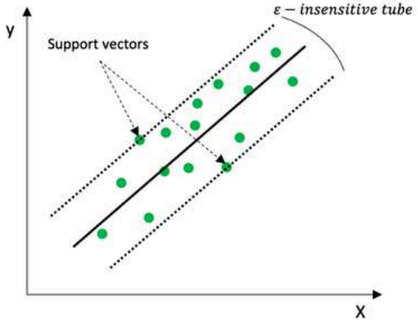
Outlook	Temp	Humidity	Windy	Hours Played
Rainy	Hot	High	False	25
Rainy	Hot	High	True	30
Overcast	Hot	High	False	48
Sunny	Mild	High	False	46
Sunny	Cool	Normal	False	62
Sunny	Cool	Normal	True	23
Overcast	Cool	Normal	True	43
Rainy	Mild	High	False	36
Rainy	Cool	Normal	False	38
Sunny	Mild	Normal	False	46
Rainy	Mild	Normal	True	48
Overcast	Mild	High	True	52
Overcast	Hot	Normal	False	44
Sunny	Mild	High	True	30



Decision Tree



Classification problem using SVM



Regression problem using SVR

Support Vector Regressor

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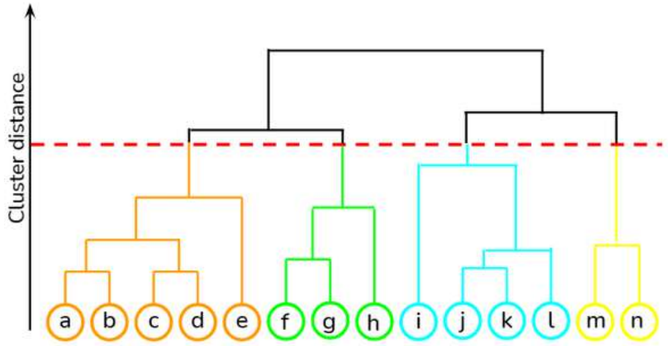
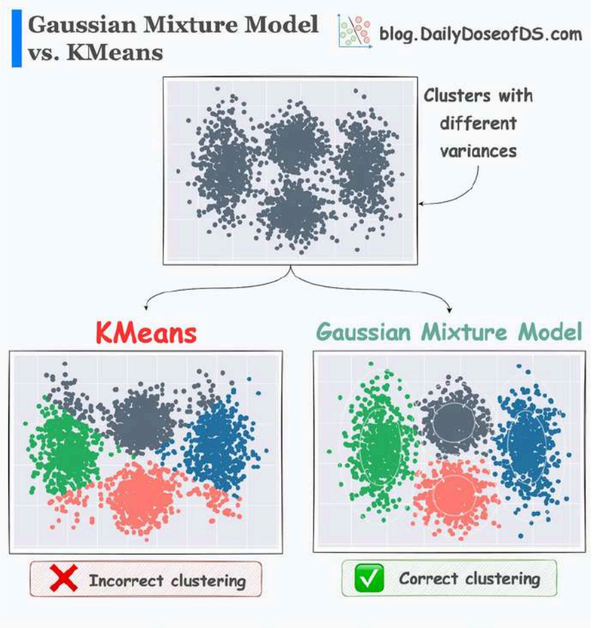
CLUSTERING

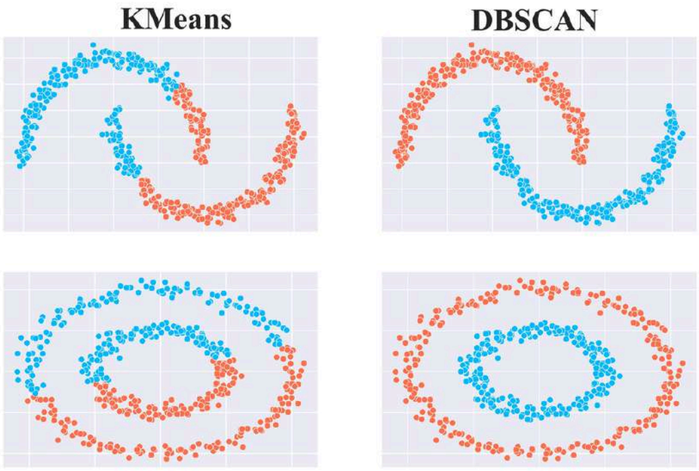
K=Means



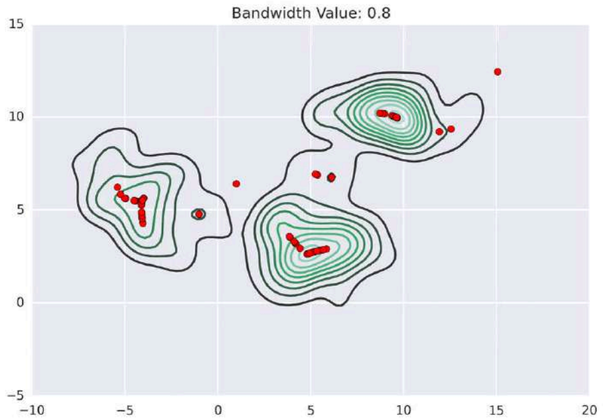
agglomerative hierarchical clustering



Gaussian Mixture



DBSCAN



Mean Shift Clustering

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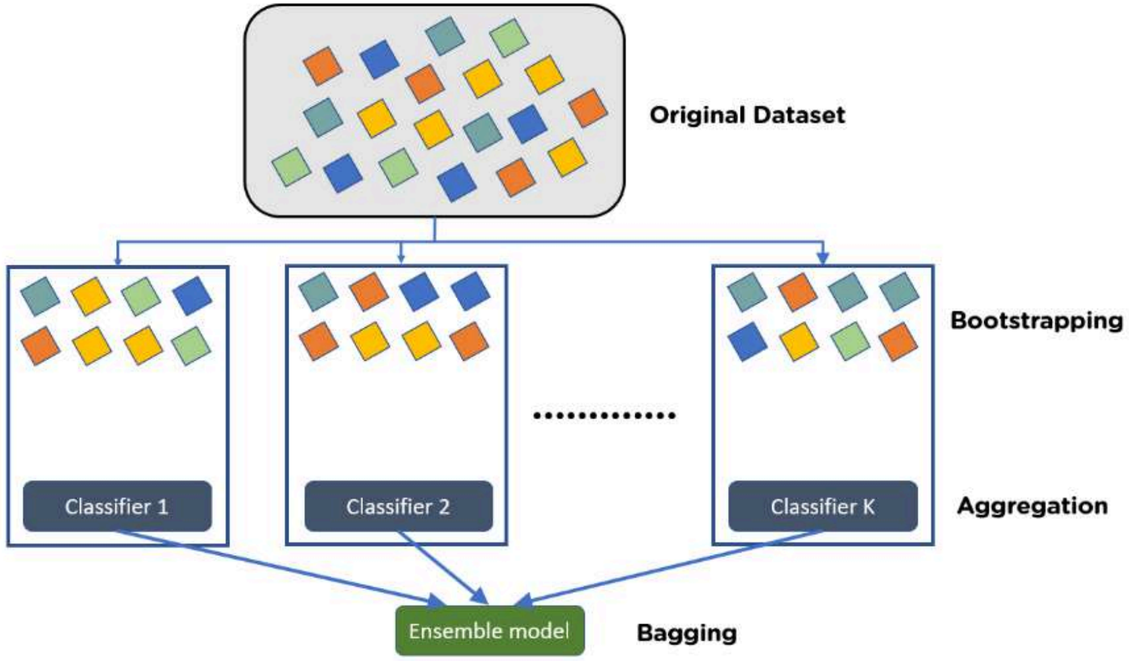
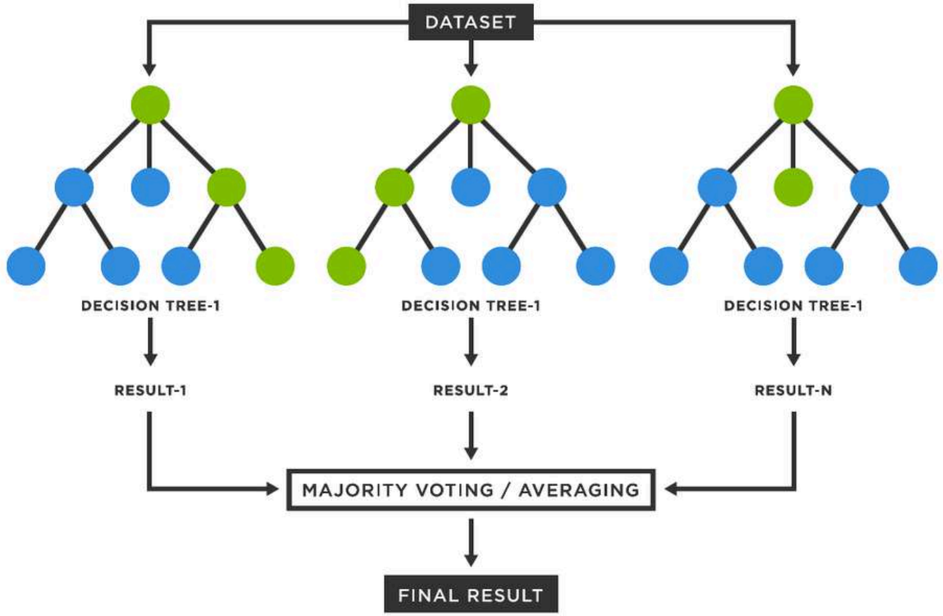
ENSEMBLE LEARNING

Ensemble learning is a technique in machine learning in which multiple models (often referred to as "base learners" or "weak learners") are combined to improve predictive performance compared to using a single model. The basic idea behind ensemble learning is that by combining multiple models that may have different weaknesses, we can produce a stronger and more stable model overall.



Bootstrap Aggregating



Random Forest

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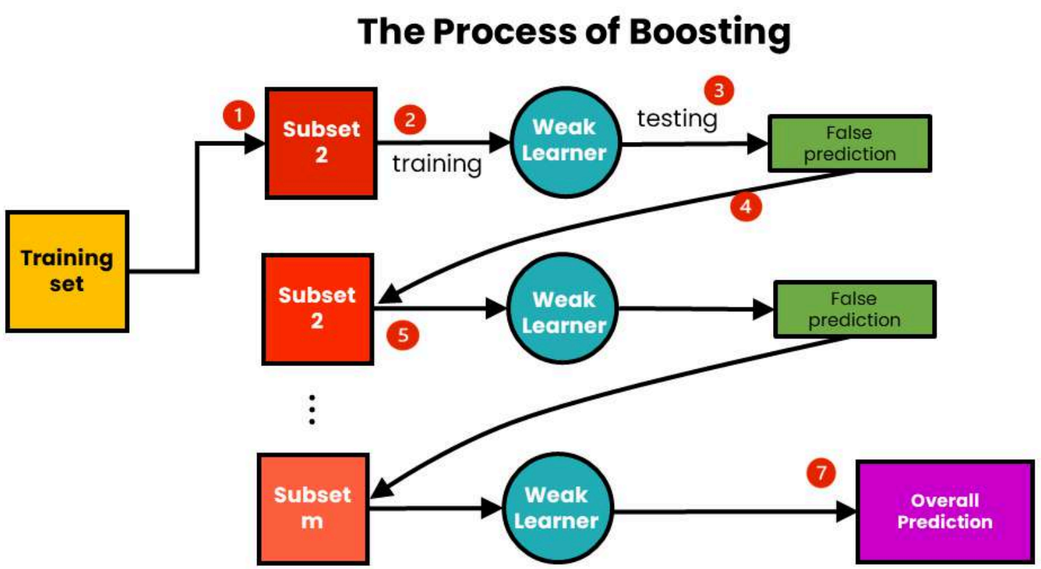
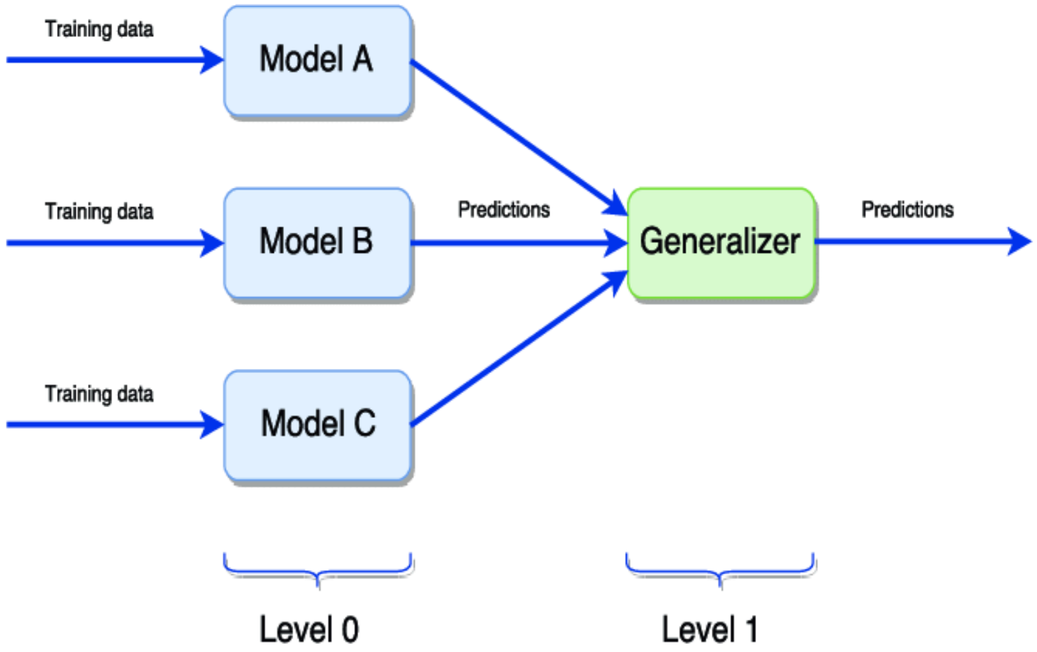


Boosting



Stacking

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ENSEMBLE LEARNING

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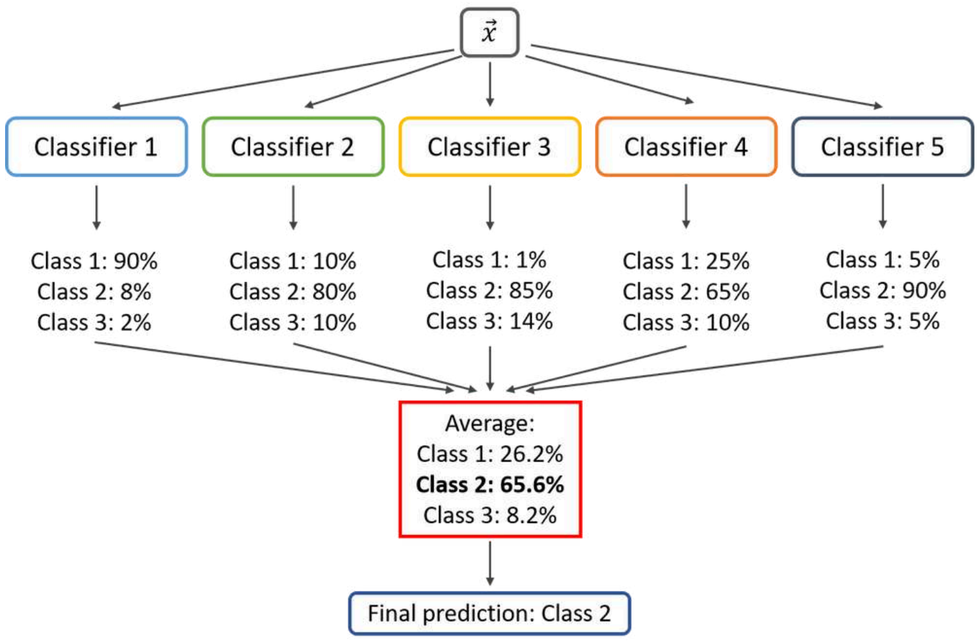
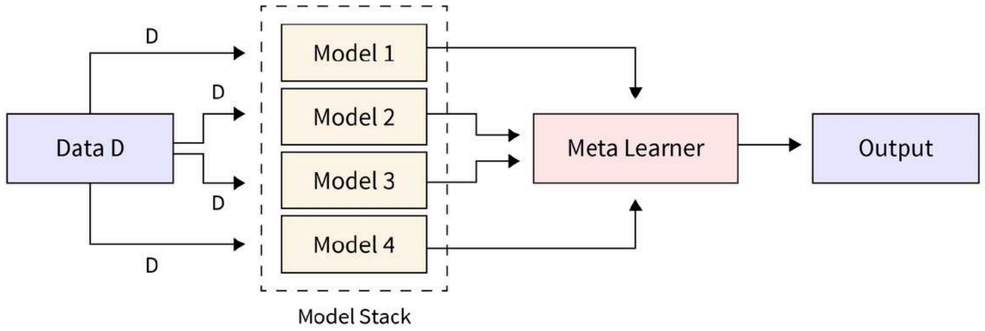


Voting



SCALER Topics

Blending

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MACHINE & DEEP LEARNING

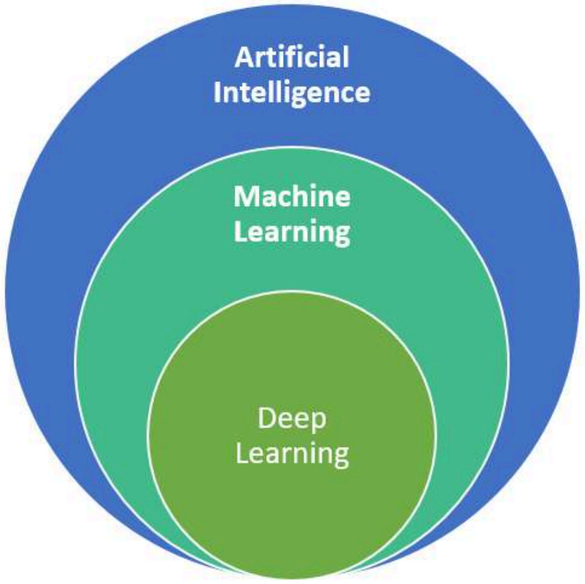
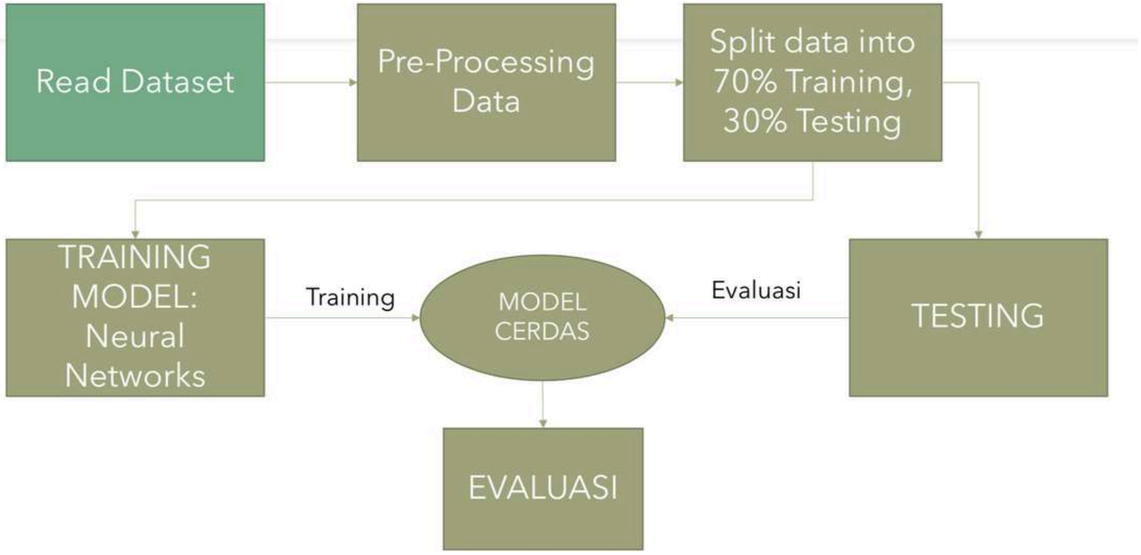


Figure 1: artificial intelligence, machine leaning and deep learning Source: Nadia BERCHANE (M2 IESCI, 2018)



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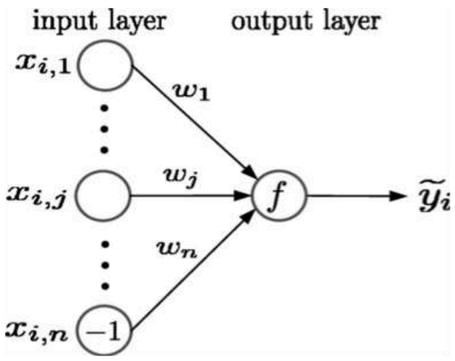
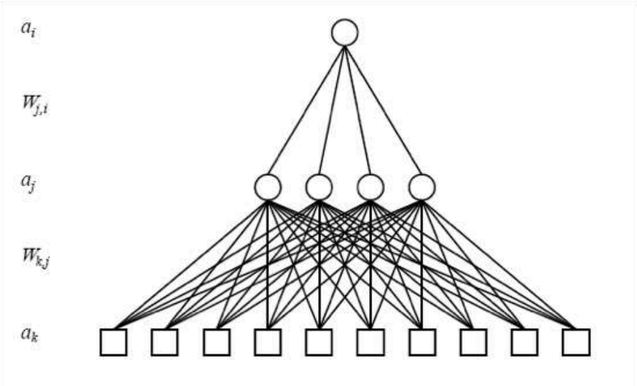
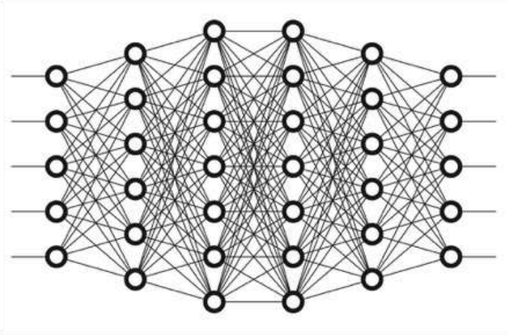
ARCHITECTURE



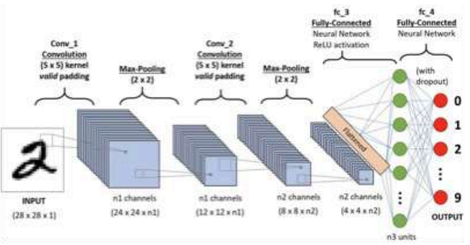
Single Layer Perceptron



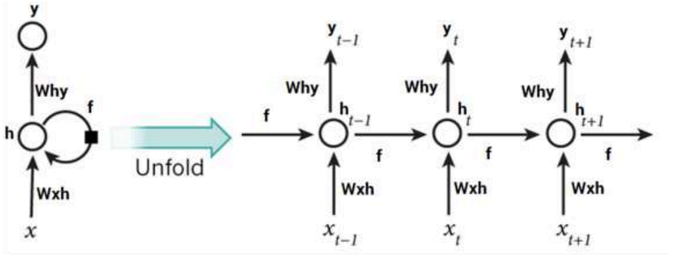
Multi Layer Perceptron



Deep Learning



Convolutional Neural Network (CNN)



Recurrent Neural Network (RNN)

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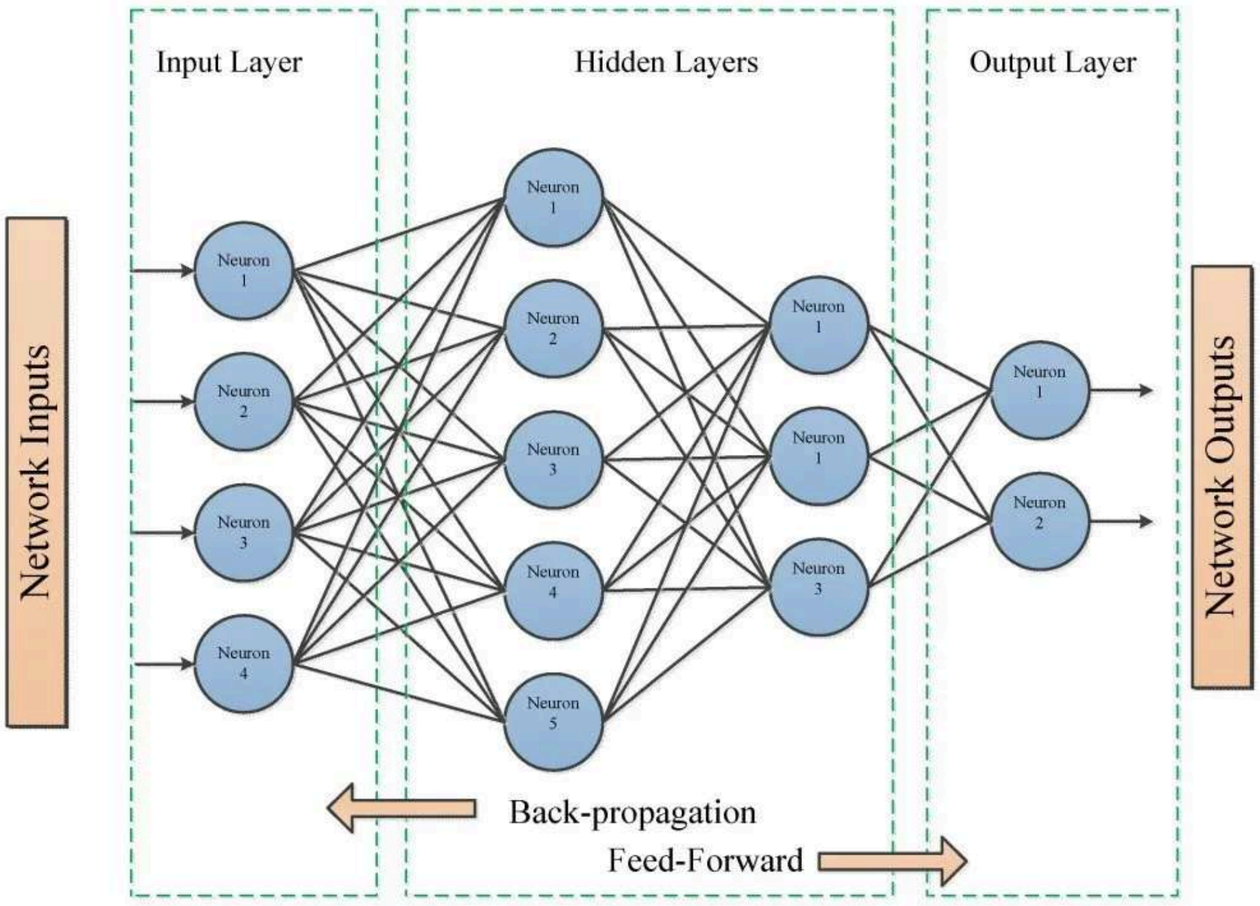
NEURAL NETWORK



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ACTIVATION FUNCTION

ReLU (Rectified Linear Unit): menggantikan nilai negatif dengan nol dan mempertahankan nilai positif.

$$f(x) = \max(0, x)$$

Sigmoid : memetakan input ke nilai antara 0 dan 1. Nilai kecil mendekati 0 dan nilai besar mendekati 1.

$$f(x) = \frac{1}{1 + e^{-x}}$$

Softmax: menghitung distribusi probabilitas untuk setiap kelas. Output dari setiap kelas dijumlahkan menjadi 1, dengan setiap kelas mendapatkan probabilitas.

$$f(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

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ACTIVATION FUNCTION

Tanh (Hyperbolic Tangent): memetakan input ke nilai antara -1 dan 1. Mirip dengan sigmoid, tetapi outputnya terpusat di sekitar nol, yang membantu mengurangi masalah saturasi.

$$f(x) = \tanh(x) = \frac{2}{1 + e^{-2x}} - 1$$

Leaky ReLU: Modifikasi dari ReLU yang membiarkan bagian negatif dari input melewati fungsi aktivasi dengan kemiringan kecil (misal 0.01).

$$f(x) = \max(0.01x, x)$$

Swish: Eksploratif dalam model deep learning besar dan state-of-the-art seperti EfficientNet.

$$f(x) = x \cdot \text{sigmoid}(x)$$

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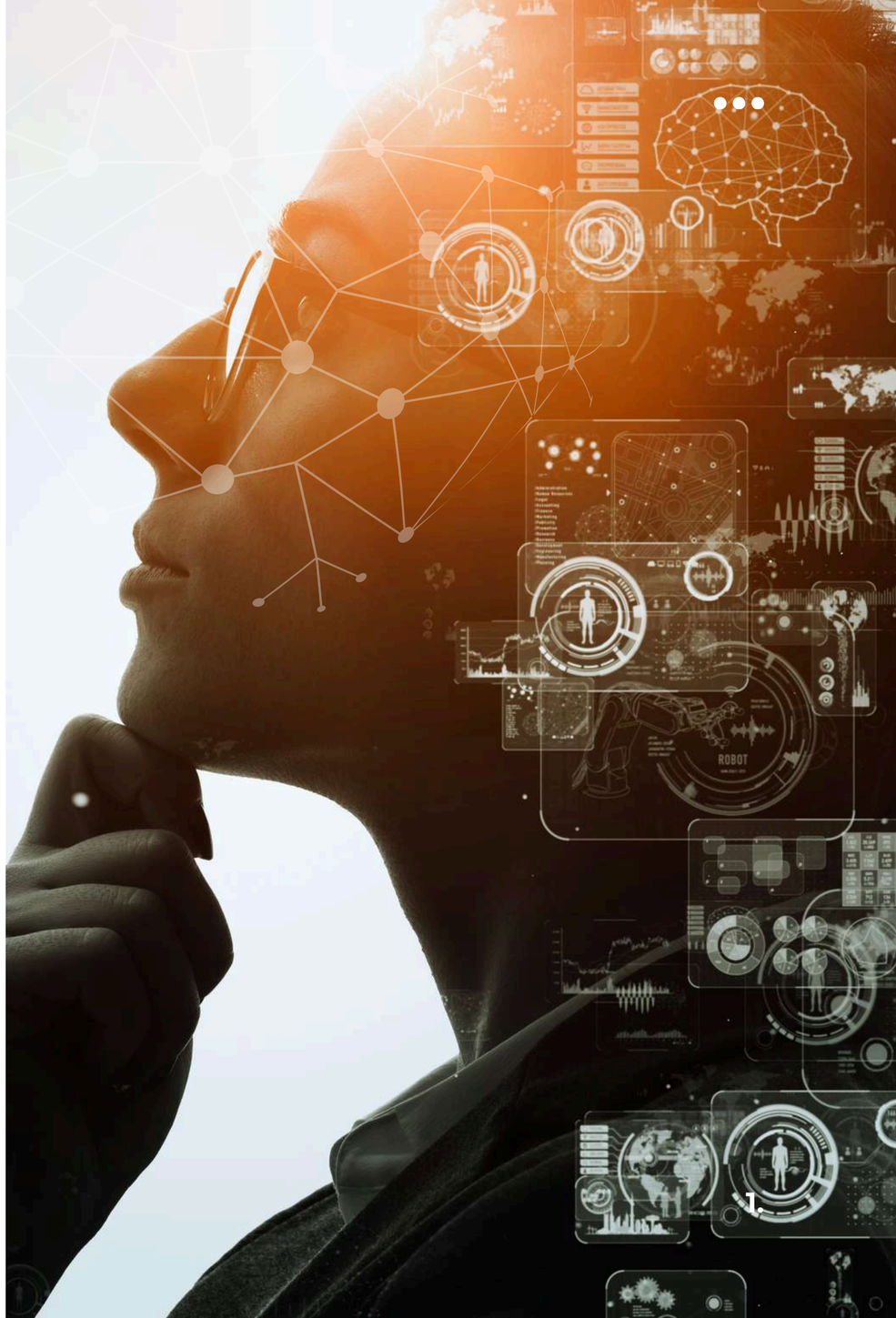
OPTIMIZER

- Stochastic Gradient Descent (SGD)
- Adam (Adaptive Moment Estimation)
- RMSProp (Root Mean Square Propagation)
- Adagrad (Adaptive Gradient Algorithm)
- Adadelta
- Momentum

THANK YOU

sariharjo01@gmail.com